

Section : Chemistry

Q.1 Consider the following statements about the physical and chemical characteristics of interstitial compounds and identify the correct statements

- (A) They have high melting points.
- (B) They are very hard.
- (C) They retain metallic conductivity.
- (D) They are chemically very reactive.

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (C) and (D) only
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.2 Arrange the following in increasing order of their molarity when their equal weights are dissolved in 250 mL of water.

- (A) glucose (180 g mol^{-1})
- (B) sodium carbonate (106 g mol^{-1})
- (C) sodium hydroxide (40 g mol^{-1})
- (D) acetic acid (60 g mol^{-1})

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (A), (D), (C), (B)
- 3. (A), (B), (D), (C)
- 4. (C), (D), (B), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.3 Isomerism shown by the complex pair $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ is _-_-_-_-_-_-

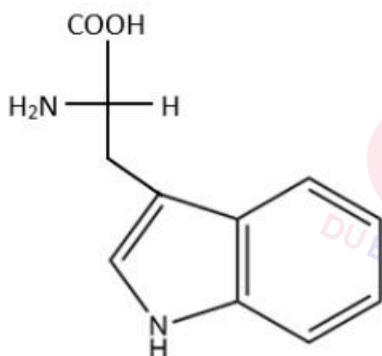
- 1. Hydrate isomerism
- 2. Coordination isomerism
- 3. Linkage isomerism
- 4. Geometrical isomerism

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.4

Identify the name of the amino acid representing the following structure



1. Phenylalanine
2. Asparagine
3. Proline
4. Tryptophan

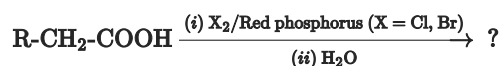
Options 1. 1

2. 2

3. 3

4. 4

Q.5 What is the product of the following reaction?



1. α -Halocarboxylic acid
2. β -Halocarboxylic acid
3. α, β -Dihalocarboxylic acid
4. 1,1-Dihalocarboxylic acid

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Arrange the following oxidation states in increasing order

- (A) characteristic oxidation state of lanthanoids
- (B) most stable oxidation state of Mn in its oxides
- (C) most stable oxidation state of Mn
- (D) oxidation state of Mn in MnO₂

Choose the correct answer from the options given below:

1. (A), (C), (B), (D)
2. (A), (B), (C), (D)
3. (C), (A), (D), (B)
4. (C), (A), (B), (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Sodium salts of carboxylic acids (RCOONa) when heated with sodalime, yield

1. R-H
2. (R-COO)₂Ca
3. RCOONa
4. RCHO

Options 1. 1

4.4

1. X: p-Hydroxyazobenzene; Y: p-Aminoazobenzene
2. X: p-Hydroxybenzene; Y: p-Aminobenzene
3. X: o-Hydroxyazobenzene; Y: o-Aminoazobenzene
4. X: p-Hydroxyazobenzene; Y: o-Aminoazobenzene

4.4

(D) $\text{C}_6\text{H}_{12}\text{O}_6$

4. (D), (B), (C), (A)

4.4

Match the catalysts given in **List-I** with the process catalyzed by them given in **List-II**

List-I	List-II
Catalyst	Process
(A) Ni in presence of Hydrogen	(I) Ziegler Natta Catalyst
(B) Finely divided iron	(II) Vegetable oil to ghee
(C) $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$	(III) Sandmeyer reaction
(D) Cu_2Cl_2	(IV) Haber's process

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4.4

Q.11

Match the complexes given in **List-I** with the geometry given in **List-II**

List-I	List-II
Complex	Geometry
(A) $[\text{NiCl}_4]^{2-}$	(I) Tetrahedral, paramagnetic
(B) $[\text{Ni}(\text{CN})_4]^{2-}$	(II) Square planer, diamagnetic
(C) $[\text{CoF}_6]^{3-}$	(III) Octahedral, diamagnetic
(D) $[\text{Co}(\text{NH}_3)_6]^{3+}$	(IV) Octahedral, paramagnetic

Choose the **correct** answer from the options given below:

- (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.12

Match the elements given in **List-I** with electronic configuration given in **List-II**

List-I	List-II
Element	Electronic configuration
(A) Cerium	(I) $[\text{Xe}] 4f^{12} 6s^2$
(B) Erbium	(II) $[\text{Rn}] 5f^7 6d^1 7s^2$
(C) Uranium	(III) $[\text{Xe}] 4f^1 5d^1 6s^2$
(D) Curium	(IV) $[\text{Rn}] 5f^3 6d^1 7s^2$

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.13 Match List-I with List-II

List-I List-II

(A) Vitamin B1 (I) Ascorbic Acid

(B) Vitamin B2 (II) Pyridoxine

(C) Vitamin B6 (III) Riboflavin

(D) Vitamin C (IV) Thiamine

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Choose the correct statement for the collision theory of chemical reactions

1. Proper orientation of the colliding particles makes them bounce back
2. If the energy of the colliding particles is greater than the threshold energy, the collision is not effective
3. Collision theory considers the atoms or the molecules as hard (solid and elastic) spheres
4. Activation energy is the only criteria for determining an effective collision

Options 1. 1

2. 2
3. 3
4. 4

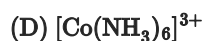
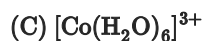
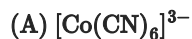
Q.15 The rate of decomposition of gaseous ammonia on a hot platinum surface at a high pressure does not change when the initial concentration of ammonia is doubled. This is an example of

1. Second order reaction
2. First order reaction
3. Zero order reaction
4. Third order reaction

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Arrange the following complexes in decreasing order of absorption of wavelength of light in the visible region:



Choose the correct option from below:

1. (B), (C), (D), (A)
2. (B), (C), (A), (D)
3. (D), (C), (B), (A)
4. (A), (D), (B), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.17



The reaction shown above is known as

1. Gabriel phthalimide synthesis
2. Gatterman reaction

3. Carbylamine reaction
4. Hoffmann bromamide degradation reaction

Options 1. 1
2. 2
3. 3
4. 4

Q.18 Among the following, which alkyl halide has the highest boiling point?

1. RI
2. RCl
3. RBr
4. RF

Options 1. 1
2. 2
3. 3
4. 4

Q.19 In the amalgam of mercury with sodium, the correct option for the physical states of solute and solvent is

1. Solute: solid and Solvent: solid
2. Solute: liquid and Solvent: liquid
3. Solute: liquid and Solvent: solid
4. Solute: solid and Solvent: liquid

Options 1. 1
2. 2
3. 3
4. 4

Q.20 The synthesis of alkyl fluoride is best achieved through

1. Swartz reaction
2. Finkelstein reaction
3. Free radical Fluorination
4. Hydrogen fluoride

Options 1. 1
2. 2
3. 3
4. 4

Q.21 Which of the following is a straight chain polysaccharide composed only of β -D-glucose units?

1. Gums
2. Starch
3. Glycogen
4. Cellulose

Options 1. 1
2. 2
3. 3
4. 4

Q.22 Name the following reaction.

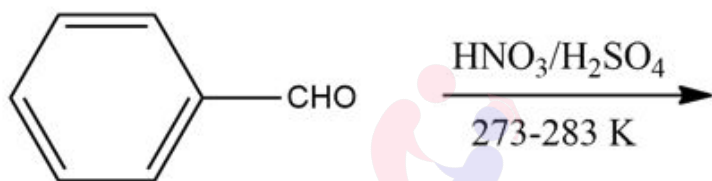


1. Gabriel synthesis
2. Sandmeyer reaction
3. Carbylamine reaction
4. Gatterman reaction

Options 1. 1
2. 2
3. 3
4. 4

Q.23

What is the product of the following reaction?



1. m-Nitrobenzaldehyde
2. p-Nitrobenzaldehyde
3. m-sulphonatobenzaldehyde
4. Benzoic acid

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Intermolecular hydrogen bonding is observed between chloroform and acetone.

Therefore, their binary solution will_

- (A) Show positive deviation from Raoult's law
- (B) Show negative deviation from Raoult's law
- (C) Obey Raoult's law
- (D) Show decrease in vapor pressure

Choose the correct answer from the options given below:

1. (A) and (D) are correct
2. (B) and (D) are correct
3. (B) and (C) are correct
4. (A) and (C) are correct

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Calculated value of magnetic moment (BM) of an aqueous solution of Mn^{2+} is.....

1. 5.92

2. 4.90

3. 3.87

4. 2.84

Options 1. 1

2. 2

3. 3

4. 4

Q.26

Match the complex given in **List-I** with the number of unpaired electrons given in **List-II**

List-I	List-II
Complex	Number of unpaired electrons (n)
(A) $[\text{NiCl}_4]^{2-}$	(I) 5
(B) $[\text{Co}(\text{NH}_3)_6]^{3+}$	(II) 1
(C) $[\text{FeF}_6]^{3-}$	(III) 2
(D) $[\text{Fe}(\text{CN})_6]^{3-}$	(IV) 0

Choose the **correct** answer from the options given below:

- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

Options 1. 1

- 2
- 3
- 4

Q.27

It was found that 0.1 mol L^{-1} of ammonia took 200 s to be decomposed to half the initial concentration. If the decomposition follows zero order reaction, then the rate constant is

- $2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- $5.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- $2 \times 10^3 \text{ mol L}^{-1} \text{ s}^{-1}$
- $2.5 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$

Options 1. 1

- 2
- 3
- 4

Q.28

Choose the statements that are valid for the molecularity of a reaction
 (A) Molecularity is equal to the number of reacting species involved in the overall reaction
 (B) Decomposition of ammonium nitrite is an example of unimolecular reaction
 (C) Reactions with the molecularity three are very rare
 (D) Molecularity can only be determined experimentally
 Choose the correct answer from the options given below:

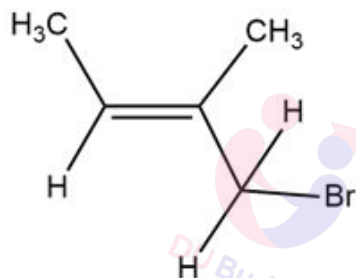
- (A), (B), (C) and (D)
- (B) and (C) only
- (A), (C) and (D) only
- (B), (C) and (D) only

Options 1. 1

- 2
- 3
- 4

Q.29

What is the IUPAC name of the following compound?



1. 1-Bromo-2-methylbut-2-ene
2. 4-Bromo-3-methylbut-2-ene
3. 3-Bromo-1,2-dimethylprop-1-ene
4. 1-Bromo-2,3-dimethylprop-2-ene

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Which of the following reagents convert an alcohol into an alkyl chloride?

- A) Conc. Hydrogen Chloride
- B) Phosphorus trichloride
- C) Phosphorus pentachloride
- D) Thionyl Chloride

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D)
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Predict which of the following will be colored in its aqueous solution?

1. Ti^{4+}
2. Sc^{3+}
3. Mn^{2+}
4. Zn^{2+}

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Consider the following statements and select the correct statements:

- (A) The energy required to form the activated complex is known as activation energy.
- (B) The Arrhenius equation explains the temperature dependence of the rate of chemical reaction.
- (C) The number of collisions per second per unit volume of the reaction mixture is known as collision frequency.
- (D) Lower the value of activation energy, lower will be the rate of reaction.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which of the following represents Fehling solution B?

1. Alkaline sodium potassium tartarate
2. Aqueous copper sulphate
3. Ammonical silver nitrate solution
4. Zn-Hg/HCl

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Milk sugar, a disaccharide, is composed of

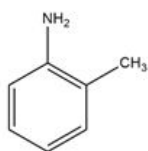
1. β -D-Galactose and β -D-Glucose
2. β -D-Galactose and α -D-Glucose
3. α -D-Glucose and β -D-Fructose
4. β -D-Glucose and α -D-Glucose

Options 1. 1

2. 2
3. 3
4. 4

Q.35

The common name and IUPAC name of the following compound, respectively, are

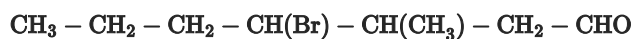


1. Aniline and 2-methylaniline
2. o-Toluidine and 2-Methylaniline
3. o-Toluidine and Benzenamine
4. m-Toluidine and Benzenamine

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Give IUPAC name of the following compound:

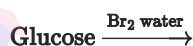


1. 3-Bromo-2-methylhexanal
2. 4-Bromo-3-methylheptanal
3. 4-Bromo-5-methylheptanal
4. 4-Bromo-3-methylheptanal

Options 1. 1

2. 2
3. 3
4. 4

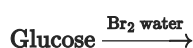
Q.37 The product of the following chemical reaction is



1. Gluconic acid
2. Saccharic acid
3. Glyceraldehyde
4. n-Hexane

Options 1. 1
2. 2
3. 3
4. 4

Q.38 The product of the following chemical reaction is



1. Gluconic acid
2. Saccharic acid
3. Glyceraldehyde
4. n-Hexane

Options 1. 1
2. 2
3. 3
4. 4

Q.39 Aniline reacts with bromine water at room temperature and forms a compound X. Identify X.

1. 2-Bromoaniline
2. 4-Bromoaniline
3. 2,4,6-tribromoaniline
4. 2-Bromoaniline and 4-Bromoaniline

Options 1. 1
2. 2
3. 3
4. 4

Q.40 If some blood cells are placed in a solution containing more than 0.9% (m/v) NaCl, then the water inside the cells will flow out of the cells and they will shrink. Such a solution is known as

1. Hypotonic
2. Hypertonic
3. Isotonic
4. Hypsotonic

Options 1. 1
2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the questions

The resistance (R) of an object is given by the expression

$$R = \rho \frac{l}{A}.$$

The inverse of R is called conductance and the inverse of ρ is called conductivity or specific conductance represented by κ .

The conductivity depends upon the nature of the material (insulator, semiconductor or conductor) and the temperature of the substance.

Similarly, the conductance due to the ions present in the solution

(ionic conductance) depends upon the nature of the electrolyte, size of ions, the nature of the solvent and the temperature.

Molar conductivity (Λ_m) of a solution at a given concentration is the conductance of volume V containing one mole of electrolyte kept between two electrodes of area A and unit distance apart.

It is given by the expression

$$\Lambda_m = \kappa V.$$

Molar conductivity increases with decrease in concentration and when concentration approaches zero, it is called limiting molar conductivity.

The decrease is linear for a strong electrolyte. However, for a weak electrolyte the change is gradual at higher concentrations and sharp near zero concentration.

Using limiting molar conductivity values of strong electrolytes we can calculate the limiting molar conductivity of weak electrolytes using Kohlrausch law of independent migration of ions.

From molar conductivity and limiting molar conductivity

we can determine the degree of dissociation $\left(\frac{\Lambda_m}{\Lambda_m^0}\right)$

and the ionization constant of a weak acid.

Select the correct statement.

1. Molar conductivity increases with the decrease in concentration for both strong and weak electrolytes.
2. Molar conductivity decreases with the decrease in concentration for both strong and weak electrolytes.
3. Molar conductivity decreases with the decrease in concentration for strong electrolytes only.
4. Molar conductivity decreases with the decrease in concentration for weak electrolytes only.

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the passage carefully and answer the questions

The resistance (R) of an object is given by the expression

$$R = \rho \frac{l}{A}.$$

The inverse of R is called conductance and the inverse of ρ is called conductivity or specific conductance represented by κ .

The conductivity depends upon the nature of the material (insulator, semiconductor or conductor) and the temperature of the substance.

Similarly, the conductance due to the ions present in the solution (ionic conductance) depends upon the nature of the electrolyte, size of ions, the nature of the solvent and the temperature.

Molar conductivity (Λ_m) of a solution at a given concentration is the conductance of volume V containing one mole of electrolyte kept between two electrodes of area A and unit distance apart.

It is given by the expression

$$\Lambda_m = \kappa V.$$

Molar conductivity increases with decrease in concentration and when concentration approaches zero, it is called limiting molar conductivity.

The decrease is linear for a strong electrolyte. However, for a weak electrolyte the change is gradual at higher concentrations and sharp near zero concentration.

Using limiting molar conductivity values of strong electrolytes we can calculate the limiting molar conductivity of weak electrolytes using Kohlrausch law of independent migration of ions.

From molar conductivity and limiting molar conductivity

we can determine the degree of dissociation $\left(\frac{\Lambda_m}{\Lambda_m^0}\right)$

and the ionization constant of a weak acid.

For a weak acid like acetic acid, the correct expression for the ionization constant at concentration c is

$$4.K_a = \frac{c \Lambda_m^2}{\Lambda_m^0 (\Lambda_m - \Lambda_m^2)}$$

4.4

The resistance (R) of an object is given by the expression

$$R = \rho \frac{l}{A}.$$

Similarly, the conductance due to the ions present in the solution (ionic conductance) depends upon the nature of the electrolyte, size of ions, the nature of the solvent and the temperature.

It is given by the expression

$$\Lambda_m = \kappa V.$$

Using limiting molar conductivity values of strong electrolytes we can calculate the limiting molar conductivity of weak electrolytes using Kohlrausch law of independent migration of ions.

we can determine the degree of dissociation $\left(\frac{\Lambda_m}{\Lambda_m^0}\right)$

SI units of conductance and conductivity, respectively, are ______

1. S and $S \text{ m}^{-1}$
2. Ω and $\Omega \text{ m}^{-1}$
3. S and $S^{-1} \text{ m}^{-1}$
4. Ω and $\Omega^{-1} \text{ m}^{-1}$

4.4

The resistance (R) of an object is given by the expression

$$R = \rho \frac{l}{A}.$$

The inverse of R is called conductance and the inverse of ρ is called conductivity or specific conductance represented by κ .

The conductivity depends upon the nature of the material (insulator, semiconductor or conductor) and the temperature of the substance. Similarly, the conductance due to the ions present in the solution (ionic conductance) depends upon the nature of the electrolyte, size of ions, the nature of the solvent and the temperature.

Molar conductivity (Λ_m) of a solution at a given concentration is the conductance of volume V containing one mole of electrolyte kept between two electrodes of area A and unit distance apart.

It is given by the expression

$$\Lambda_m = \kappa V.$$

Molar conductivity increases with decrease in concentration and when concentration approaches zero, it is called limiting molar conductivity.

The decrease is linear for a strong electrolyte. However, for a weak electrolyte the change is gradual at higher concentrations and sharp near zero concentration.

Using limiting molar conductivity values of strong electrolytes we can calculate the limiting molar conductivity of weak electrolytes using Kohlrausch law of independent migration of ions.

From molar conductivity and limiting molar conductivity

we can determine the degree of dissociation $\left(\frac{\Lambda_m}{\Lambda_m^0}\right)$

and the ionization constant of a weak acid.

Which of the following is an insulator?

1. Graphite
2. Teflon
3. Germanium
4. Sodium

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the passage carefully and answer the questions

The resistance (R) of an object is given by the expression

$$R = \rho \frac{l}{A}.$$

The inverse of R is called conductance and the inverse of ρ is called conductivity or specific conductance represented by κ .

The conductivity depends upon the nature of the material (insulator, semiconductor or conductor) and the temperature of the substance.

Similarly, the conductance due to the ions present in the solution (ionic conductance) depends upon the nature of the electrolyte, size of ions, the nature of the solvent and the temperature.

Molar conductivity (Λ_m) of a solution at a given concentration is the conductance of volume V containing one mole of electrolyte kept between two electrodes of area A and unit distance apart.

It is given by the expression

$$\Lambda_m = \kappa V.$$

Molar conductivity increases with decrease in concentration and when concentration approaches zero, it is called limiting molar conductivity.

The decrease is linear for a strong electrolyte. However, for a weak electrolyte the change is gradual at higher concentrations and sharp near zero concentration.

Using limiting molar conductivity values of strong electrolytes we can calculate the limiting molar conductivity of weak electrolytes using Kohlrausch law of independent migration of ions.

From molar conductivity and limiting molar conductivity

we can determine the degree of dissociation $\left(\frac{\Lambda_m}{\Lambda_m^0}\right)$

and the ionization constant of a weak acid.

The limiting molar conductivity values for HCl , NaCl and CH_3COONa are 425.9, 126.4 and 91.0 $\text{S cm}^2 \text{mol}^{-1}$, respectively.

Calculate the limiting molar conductivity value for acetic acid.

1. $643.3 \text{ S cm}^2 \text{ mol}^{-1}$
2. $208.5 \text{ S cm}^2 \text{ mol}^{-1}$
3. $390.5 \text{ S cm}^2 \text{ mol}^{-1}$
4. $461.3 \text{ S cm}^2 \text{ mol}^{-1}$

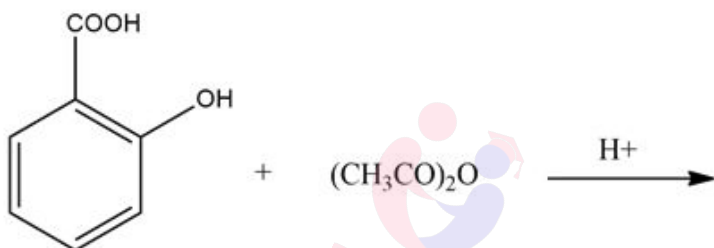
Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the questions

Alcohols and phenols are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively is replaced by -OH group. These are further classified on the basis of number of hydroxy groups and according to the hybridization of the carbon atom sp^2 or sp^3 to which the -OH group is attached. They find applications in industry as well as in day-to-day life. One of the most commercially obtained alcohol also known as wood spirit is obtained by destructive distillation of wood. Both alcohols and phenols are acidic in nature. The acidity of alcohols and phenols vary according to the nature of substituent(s) attached to them. Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters. Introduction of acetyl group in alcohols and phenols is called acetylation. Phenol undergo electrophilic aromatic substitution reaction because the -OH activates the benzene ring. It undergoes nitration, halogenation and sulphonation reactions.

The final product of the following reaction is:



1. Paracetamol
2. Ibuprofen
3. Aspirin
4. Salicylaldehyde

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the questions

Alcohols and phenols are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively is replaced by -OH group. These are further classified on the basis of number of hydroxy groups and according to the hybridization of the carbon atom sp^2 or sp^3 to which the -OH group is attached. They find applications in industry as well as in day-to-day life. One of the most commercially obtained alcohol also known as wood spirit is obtained by destructive distillation of wood. Both alcohols and phenols are acidic in nature. The acidity of alcohols and phenols vary according to the nature of substituent(s) attached to them. Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters. Introduction of acetyl group in alcohols and phenols is called acetylation. Phenol undergo electrophilic aromatic substitution reaction because the -OH activates the benzene ring. It undergoes nitration, halogenation and sulphonation reactions.

Which of the following compound lacks a phenolic group?

1. Quinol

2. Glycerol
3. Resorcinol
4. Cresol

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the questions

Alcohols and phenols are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively is replaced by -OH group. These are further classified on the basis of number of hydroxy groups and according to the hybridization of the carbon atom sp^2 or sp^3 to which the -OH group is attached. They find applications in industry as well as in day-to-day life. One of the most commercially obtained alcohol also known as wood spirit is obtained by destructive distillation of wood. Both alcohols and phenols are acidic in nature. The acidity of alcohols and phenols vary according to the nature of substituent(s) attached to them. Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters. Introduction of acetyl group in alcohols and phenols is called acetylation. Phenol undergo electrophilic aromatic substitution reaction because the -OH activates the benzene ring. It undergoes nitration, halogenation and sulphonation reactions.

Which compound is also known as 'wood spirit'?

1. Ethanol
2. Methanol
3. Resorcinol
4. Phenol

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions

Alcohols and phenols are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively is replaced by -OH group. These are further classified on the basis of number of hydroxy groups and according to the hybridization of the carbon atom sp^2 or sp^3 to which the -OH group is attached. They find applications in industry as well as in day-to-day life. One of the most commercially obtained alcohol also known as wood spirit is obtained by destructive distillation of wood. Both alcohols and phenols are acidic in nature. The acidity of alcohols and phenols vary according to the nature of substituent(s) attached to them. Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters. Introduction of acetyl group in alcohols and phenols is called acetylation. Phenol undergo electrophilic aromatic substitution reaction because the -OH activates the benzene ring. It undergoes nitration, halogenation and sulphonation reactions.

When phenol is treated with concentrated sulphuric acid, then it gets converted into phenol-2,4-disulphonic acid (X). On further treatment of X with conc. nitric acid, product Y is formed. Identify Y.

1. Salicylic acid
2. Picric acid
3. o-Nitrophenol
4. 2,4-dinitrophenol

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions

Alcohols and phenols are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively is replaced by -OH group. These are further classified on the basis of number of hydroxy groups and according to the hybridization of the carbon atom sp^2 or sp^3 to which the -OH group is attached. They find applications in industry as well as in day-to-day life. One of the most commercially obtained alcohol also known as wood spirit is obtained by destructive distillation of wood. Both alcohols and phenols are acidic in nature. The acidity of alcohols and phenols vary according to the nature of substituent(s) attached to them. Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters. Introduction of acetyl group in alcohols and phenols is called acetylation. Phenol undergo electrophilic aromatic substitution reaction because the -OH activates the benzene ring. It undergoes nitration, halogenation and sulphonation reactions.

Identify the strongest acidic compound

1. Phenol
2. p-Nitrophenol

3. p-Cresol

4. p-Methoxyphenol

Options 1. 1

2. 2

3. 3

4. 4

